DeFi & Advanced Crypto

Module 2 Study Guide

Smart Contracts & dApps

This guide covers every lesson of the module, using only words and figures from the lessons themselves. Work through it after watching the lessons, then use the checklist at the end to test yourself.

Lesson 1: How Smart Contracts Execute

In this lesson: Defining Smart Contracts; Code vs Traditional Contracts; Worked TT$ Lending Ex

Key points

- Smart contracts store rules directly on a blockchain
- Execution triggers automatically when exact conditions are met
- Transactions settle instantly without intermediaries or extra fees

Examples from the lesson

- This is Module 2, "Smart Contracts & dApps", and today we're diving into Lesson 1: "How Smart Contracts Execute".
- You want to borrow TT$ 1,000 to purchase inventory for your local market stall, but you don't want to sell your existing crypto assets.
- You agree to put up TT$ 1,500 worth of crypto collateral into a smart contract to secure the TT$ 1,000 loan.
- The smart contract holds your TT$ 1,500 collateral in a secure digital vault and immediately releases the TT$ 1,000 loan to your wallet.
- The agreement rules are simple and coded upfront: when you repay the TT$ 1,000 principal plus a TT$ 50 interest fee, the contract automatically unlocks and returns your TT$ 1,500 collateral.
- When you transfer TT$ 1,050 back into the protocol, the smart contract detects the incoming payment instantly.
- It verifies that TT$ 1,000 covers the original principal and TT$ 50 covers the fee, totaling TT$ 1,050 in settlement.
- The moment that full balance arrives, the contract automatically triggers the release of your TT$ 1,500 collateral back to your wallet.
- In our next lesson, Lesson 2: "Using dApps Safely, Step by Step", we'll look at how to interact with these contracts securely through decentralized applications without falling victim to common scams or faulty permissions.

Lesson 2: Using dApps Safely, Step by Step

In this lesson: Wallet Connection vs Permissions; Token Approvals Explained; Worked Example

Key points

- Wallet connection reveals address while approvals authorize token spending
- Always replace unlimited defaults with exact custom spending caps
- Regularly use revocation tools to close past contract permissions

Examples from the lesson

- I am so glad you joined us for the DeFi and Advanced Crypto course, Module 2: Smart Contracts and dApps.
- Today, in Lesson 2, titled Using dApps Safely, Step by Step, we are going to learn how to connect your wallet, inspect permissions, and revoke token approvals you no longer need.
- You decide to spend TT$ 1,000 worth of tokens on this transaction.
- When the dApp prompts you, it might ask for an unlimited approval by default, but you carefully change that setting to a custom allowance of exactly TT$ 1,000 worth of tokens.
- When the smart contract executes your trade, it deducts the TT$ 1,000 allowance to complete the token swap.
- That leaves you with a remaining token allowance of TT$ 0 for that specific transaction.
- However, if you had accidentally granted an unlimited allowance of TT$ 5,000 worth of tokens instead, the smart contract would still have an active permission to spend your remaining TT$ 4,000 worth of tokens at any point in the future.
- Safely navigating dApps requires a simple three-step routine every single time you interact with decentralized finance platforms.
- Step one is connecting your wallet carefully and verifying that you are on the legitimate, official website URL.
- Step two is reviewing every approval prompt in your wallet interface, ensuring you set custom spending limits rather than default unlimited allowances.
- Step three is regularly visiting a revocation tool, such as Revoke.cash or your wallet security dashboard, to cut off contract access you no longer use.
- Taking thirty seconds to clean up old approvals completely removes vector risks from past dApp interactions.
- In our next lesson, Lesson 3: Spotting Contract Risk and Audits, we will learn how to evaluate smart contract code safety and read audit reports before interacting with new platforms.

Lesson 3: Spotting Contract Risk and Audits

In this lesson: Purpose of smart contract audits; Identifying critical vulnerability ratings; Spott

Key points

- Read smart contract security audit reports carefully
- Identified critical vulnerabilities and admin key control risks
- Calculated total capital exposure to decide when to walk away

Examples from the lesson

- We are inside Module 2 on Smart Contracts and dApps, and today we are tackling Lesson 3: Spotting Contract Risk and Audits.
- If one single person holds an admin key that can drain funds in five seconds, you are not using a decentralized protocol—you are trusting a stranger with your wallet.
- Imagine you have TT$5,000 saved from your monthly salary, and you want to lock it into a yield contract to earn passive interest.
- The audit report reveals three major findings that paint a clear picture of the contract's overall health.
- First, it identifies TT$0 in unaddressed critical bugs because the team patched every single high-severity exploit during the review phase.
- You calculate the capital put at immediate risk by this single admin key feature, which equals the full TT$5,000 of your principal deposit.
- Third, the audit flags a potential transaction tax flaw that could lock up to 10 percent of your balance upon exit, representing an additional exposure of TT$500.
- Adding the TT$5,000 admin risk to the TT$500 exit risk brings your total capital exposed to potential contract risk to TT$5,500.
- Facing a total risk profile of TT$5,500 on a TT$5,000 principal deposit, you make the smart financial move and walk away immediately.
- Spotting these red flags saved your TT$5,000 from being trapped in an unsafe protocol.
- You do not need to be a software engineer to read an audit summary table or check if emergency pause powers have a 48-hour timelock.

Recap checklist — can you say each one?

Lesson 1: How Smart Contracts Execute

- ☐ Smart contracts store rules directly on a blockchain
- ☐ Execution triggers automatically when exact conditions are met
- ☐ Transactions settle instantly without intermediaries or extra fees

Lesson 2: Using dApps Safely, Step by Step

- ☐ Wallet connection reveals address while approvals authorize token spending
- ☐ Always replace unlimited defaults with exact custom spending caps
- ☐ Regularly use revocation tools to close past contract permissions

Lesson 3: Spotting Contract Risk and Audits

- ☐ Read smart contract security audit reports carefully
- ☐ Identified critical vulnerabilities and admin key control risks
- ☐ Calculated total capital exposure to decide when to walk away